gravitational pull will drop only by a factor of 2 (not 4). If the universe had only two space dimensions, then, gravitational pull would be inversely proportional to separation, not the square of separation.

If the universe had only one space dimension, as in Figure 13.5b, the law of gravity would be simpler still. Gravitational field lines would have no room to spread out, and so the force of gravity would not decrease with separation. If you were to double the distance between the sun and the

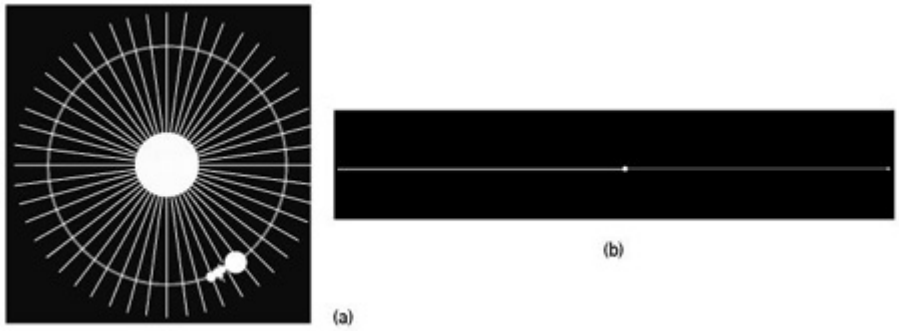


Figure 13.5 **(**a**)** In a universe with only two spatial dimensions, the gravitational force drops in proportion to separation, because gravitational field lines uniformly spread on a circle whose circumference is proportional to its radius. **(**b**)** In a universe with one space dimension, gravitational field lines do not have any room to spread, so the gravitational force is constant, regardless of separation.

satellite (assuming that versions of such objects could exist in such a universe), the same number of field lines would penetrate the satellite and hence the force of gravity acting between them would not change at all.

Although it is impossible to draw, the pattern illustrated by Figures 13.4 and 13.5 extends directly to a universe with four or five or six or any number of space dimensions. The more space dimensions there are, the more room gravitational lines of force have to spread out. And the more they spread out, the more precipitously the force of gravity drops with increasing separation. In four space dimensions, Newton’s law would be an inverse cube law (double the separation, force drops by a factor of 8); in five space dimensions, it would be an inverse fourth-power law (double the separation, force drops by a factor of 16); in six